PCC Processing Design code – Theoretical Manual

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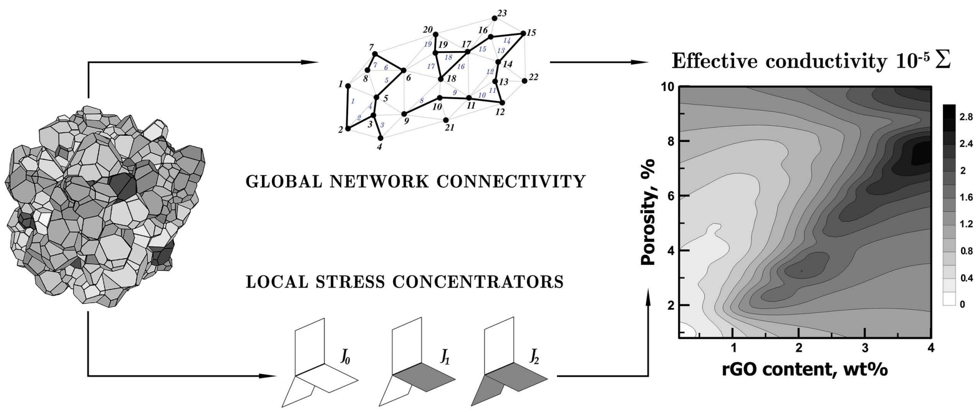
Abstract

The paper presents a theoretical manual for the background discrete mathematical tools and terminology necessary for a comprehensive understanding of the algorithms and concepts on which the Polytopal Cell Complex (PCC) Processing Design C++ code is built. It is based on the series of publications 2019-2025 years, including Borodin et al. (2019); Borodin and Jivkov (2019); Zhu et al. (2021); Borodin et al. (2021); Zhu et al. (2023a,b, 2024); Borodin et al. (2024d,c); Bushuev et al. (2025); Borodin et al. (2024b); Barroso et al. (2025).



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Contents

List of the used abbreviations and model variables

PCC	polytopal cell complex
GB	grain boundary
TJ	triple junction
QP	quadruple point
σ_p	p-cell of a PCC
$\mathcal{T}$	tessellation of a 3-dimensional space by polyhedra
$\mathcal{M}$	PCC based on a tessellation $\mathcal{T}$
$\mathcal{M}_p$	p-skeleton of a PCC

$q,\ p,\ f$	fractions of ordinary, special assigned and special generated cell types, respectively
$p_a,\ f_a$	fraction of special assigned and special generated cells containing agglomerations
D_k	triple junction d-type corresponding to $k \in \{1,2,3\}$ incident cells of the same type in the network of special cells
$d_k^a,\ d_k^g$	fraction of k-type triple junctions incident to special cells
$B_p^a,\ Q_p^a$	$(p-1)$-cell and $(p+1)$-cell indices of special assigned cells, respectively
$B_p^g,\ Q_p^g$	$(p-1)$-cell and $(p+1)$-cell indices of special generated cells, respectively
$S_{conf}^a,\ S_{conf}^g$	TJ configuration entropies of special assigned and generated cells, respectively
$\widehat{\partial}_p,\ \widehat{\delta}_p$	incidence matrices (boundary operators) of a PCC's p-skeletons and their transposed counterparts
$\partial_k,\ \delta_k$	reduced incidence matrices (boundary operators) of a PCC's special p-cells and their transposed counterparts
$\widehat{L_p},\ L_p$	full and reduced p-th combinatorial Laplacian of a PCC
W_p,	topological p-cell weight

Introduction

The present Theoretical Manual provides a necessary theoretical minimum required as a foundation for the comprehensive understanding of the numerical algorithms and concepts implemented in the PCC Processing Design (C++) and PCC Analyser (Python) codes Borodin et al. (2024a) created as a part of the MATERiA software project Borodin (2022).

The core of the methodology is employment of the fully discrete 3-dimensional (3D) and 2-dimensional (2D) representations of bulk polycrystalline structures Zhu et al. (2023b, 2021); Borodin and Jivkov (2019); Borodin et al. (2021) as a collection of abstract cells of different dimensions $p \leq 3$, where every p-cell σ_p has a boundary formed by incident $(p-1)$-cells σ_{p-1} belonging to the complex Kozlov (2008); Grady and Polimeni (2010). Such a definition makes the complex an extension of the concept of graphs Bollobas (1998) (1-complex) to a higher-dimensional case. The specific geometric realisation of the discrete complexes useful for materials science Zhu et al. (2023a,b); Bushuev et al. (2025), physics and mechanics of materials Borodin et al. (2024c); Zhu et al. (2024) and materials design Zhu et al. (2021); Borodin et al. (2021, 2024d) can be obtained as a polytopal cell complex Kozlov (2008) (PCC) associating 3-cells with the volumetric polytopes (grains) Quey (2021), 2-cells with interfaces (boundaries), 1-cells with boundary junctions and 0-cells with points. In particular, it is possible to build PCCs corresponding to

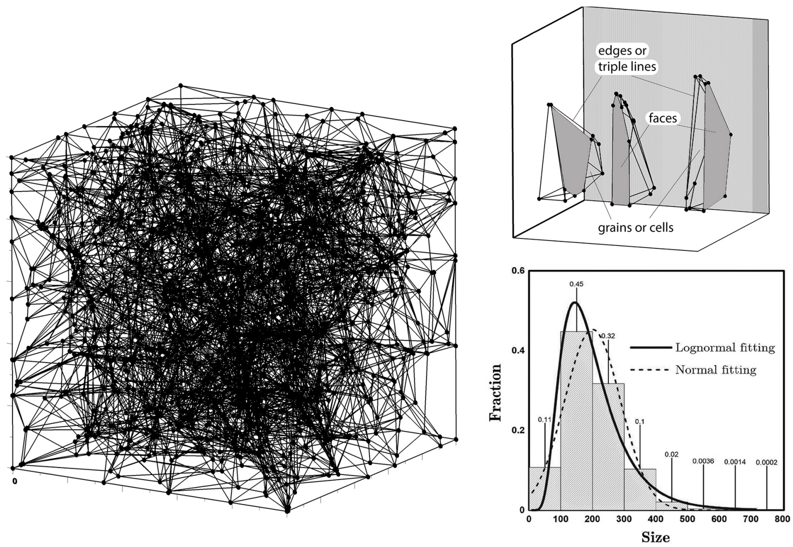

Figure 1: Voronoi tessellation of a cube constructed by Voro++ Rycroft (2009) with its three polyhedra, their low-dimensional cells and size distribution of the polyhedra in the tessellation.

the experimental data obtained by the widely used Electron Backscattering Diffraction (EBSD) Microscopy Wright et al. (2011). In that case, all cells are also additionally equipped with coordinates, sizes and orientations similar to 2D/3D EBSD maps Wright et al. (2011).

Any additional material microstructure, such as a network of special, high-angle grain boundaries Borodin and Jivkov (2019) or graphene inclusions Borodin et al. (2021) can be assigned on the corresponding PCC cells as a 1-dimensional (1D) graph structure (system of 2-sets Bollobas (1998)) with the corresponding cell colouring procedure Bollobas (1998). Hence, a large set of physical and engineering problems, including the conductive inclusion networks design Borodin et al. (2021) or multiscale fracture simulations Borodin et al. (2024d) in polycrystalline materials, can be explicitly studied using the formalism of PCCs and their special cells.

Section 1 of this Theoretical Manual introduces the basics of the more or less conventional combinatorial notions based on the mathematical theories of sets and their systems (including graph and network theories) with the ideas of labelling (colouring) and indexing of their elements. The following Section 2, Section 3, and Section 4 introduce the *polytopal cell complexes* as fully discrete (without continuum parts) combinatorial spaces possessed with their *skeletons*, combinatorial *measures*, configuration spaces, boundary operators, combinatorial Laplacians and particular cell *topologies*.

The interested reader can also find much more information and application examples in the closing *Further Reading* 6.2.1 list of suggested books, publications, codes and online resources.

1. Set Systems and Graphs

1.1. Combinatorial representation

The basis of any discrete methodology is the combinatorics of finite sets and their families Cunningham (2016), allowing revisiting of the conventional continuum approaches using tools provided by graph theory Bollobas (1998) and algebraic topology Kozlov (2008); Grady and Polimeni (2010). Fig.2 shows a single element, a set of similar elements and a family of sets. An

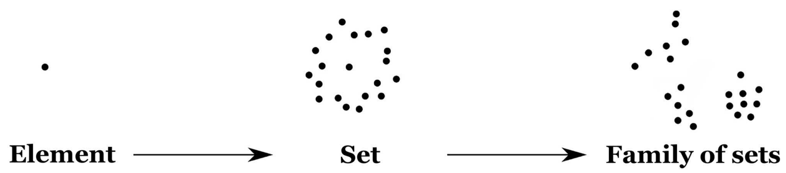

Figure 2: One element, a finite set of elements and a family of finite sets.

equivalent numerical representation can be written as:

$$\{1\} \rightarrow \{1, 2, 3, 4, ..\} \rightarrow \{\{1, 2, .., 7\}, \{8, 9, .., 13\}, \{14, .., 22\}\}.$$

Then, a *partially ordered set* (POSET) can be created by introducing a partial ordering for these sets and their elements: for some of the sets, the operators $<$, $>$, and $=$ can be defined. Let us define the *order parameter* as the size of a set, such that set X is greater ($>$) than a set Y if and only if ('iff') $|X| > |Y|$ and $Y \in X$ (set X contains the set Y). A set X *covers* another set Y iff $X > Y$ and there is no set Z, such that $X > Z > Y$. Then the *rank* function $r \in S = \{0, 1, 2, 3, ..\}$ can be introduced, such that $r(Y) = r(X) + 1$ whenever set Y covers set X. Then, *level p-sets* are all sets $X : r(X) = p$.

All these definitions allow one to define a meaningful notion for algebraic topology and its applications – *p-skeleton*, which is defined as all sets possessing the same rank p. The examples of the corresponding skeletons with their ranks are:

- Set $\{1, 2, 3, 4, 5, 6, 7, 8..\}$ with $r = 1$,
- Graph $\{1, 2\}, \{3, 4\}, \{1, 6\}, \{4, 5\}..$ with $r = 2$,
- Hypergraph $\{1, 2, 6\}, \{3, 4, 5\}, \{2, 7, 8\}..$ with $r = 3$.

Fig.3 shows all p-skeletons of a single polyhedron represented as a combinatorial complex.

Topology is another basic concept of combinatorial topology related to the specific organisation which a family of discrete sets can have. A set X along

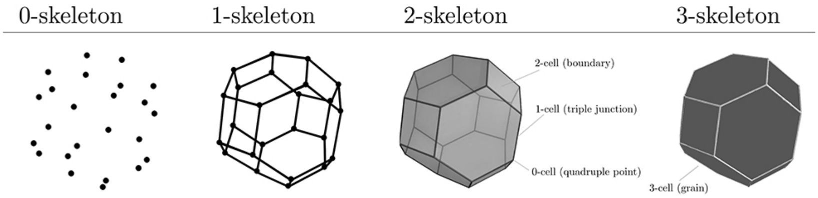

Figure 3: A single polyhedron represented as a combinatorial complex with all its p-skeletons of different ranks $p \in \{0, 1, 2, 3\}$.

with a collection X_T of its subsets is said to be a topology if the subsets in X_T obey the following conditions Bishop and Goldberg (1980):

- The (trivial) subsets X and the empty set O are in X_T.
- Whenever two sets Y and U are in X_T, then so is their intersection $Y \cap U \in X_T$.
- Whenever two or more sets $Y, U, V, ..$ are in X_T, then so is their union $Y \cup U \cup V \cup .. \in X_T$.

In terms of cell complexes Grady and Polimeni (2010), it requires that for any two p-cells, all $(p-1)$-cells on their boundaries and $(p+1)$-cells that contain them (on their co-boundary) must also belong to the same PCC that will be discussed in detail in Section 2.

The integer numbers can naturally enumerate all the set elements as any countable set is isomorphic to the set of natural numbers $\mathbb{N}$. Additionally to their *identification numbers* (IDs), each element of a set and each set in the set family (such as a PCC or its skeletons) can be endowed with *variables* possessed their specific values (Table 1), be *labelled* (coloured) with a special type (Fig.4) and *indexed*, according to some specific rules.

Sets	$\{1, 2, 3, 4, 5, 6, 7\}$	$\{8, 9, 10, 11, 12, 13\}$
Set IDs	1	2
Variables	$\{x_1, y_1, z_1, ..\}$	$\{x_2, y_2, z_2, ..\}$
Labels/Indices	$\{\#r, \#r, \#r, ..\}$	$\{\#g, \#g, \#g, ..\}$

Table 1: Sets IDs with the variables and labels (colours) assigned to their elements (cells).

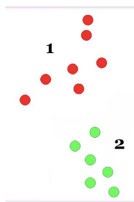

Figure 4: Two coloured sets.

Let us clearly distinguish between these categories, which is useful for applications and, especially convenient as a formalism in programming codes:

- Variables can be scalar (integers or decimal numbers) or vector-valued. Use of the normalised 'fraction' variables $f \in [0, 1]$ is especially convenient in many practical tasks;
- Labels (or colours) are cell types either *assigned* according to some rules or *generated* as the result of a kinetic process;
- Indices are cell types imposed by the existing labels (often induced by higher-dimensional cell types) according to some specific rules.

All labels can be encoded in the DNA-like sequence Mitchell (2009) referred to as a *state vectors* $S = \{r, r, g, b, \dots g, r\}$ with the length equal to the size N of the corresponding set $\{1, 2, 3, 4, \dots N-1, N\}$. Fig.5 shows the coloured cell structure and the corresponding graph whose colouring can be represented as a state vector. Every state vector can be characterised by

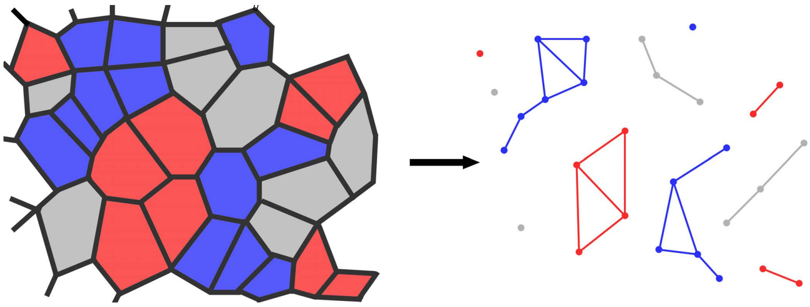

Figure 5: A sketch of a coloured structure of 2-cells and the corresponding graph.

introducing a combinatorial *measure*, such as fractions Borodin et al. (2021) and configuration entropy Frary and Schuh (2005); Borodin and Jivkov (2019). Any two state vectors can be compared using the *combinatorial distances*, such as Hellinger distance Kitsos and Toulias (2017) or Kullback–Leibler divergence MacKay (2003). Such measures will be discussed in detail further in Section 4.2.

Finally, several definitions are essential for understanding the functioning of the CPD code.

- Let us refer to the *PCC structure* or a *PCC state* as a collection of all its state vectors.
- By definition, a *process* is a sequence of related structures obtained as the result of the PCC "processing" procedure.
- Finally, *design* is the corresponding sequence of related structures created as the result of the PCC "design" optimisation procedure.

- In full analogy, a *p-process* and a *p-design* are the corresponding sequences of related structures for a particular p-skeleton of a PCC.

1.2. Algebraic representation

The combinatorial representation in the form of sets and their systems is often very convenient and widely used – for instance, in the Neper software output *.*tess* files . However, it is often less efficient for the description of connections between the elements of different ranks. Algebraic representations of graphs and networks in the form of matrices are much more efficient computationally when the neighbour (adjacent) elements and lower-dimensional boundaries (incident elements) are of prime interest. Graph and network structures contain only nodes (0-dimension cells) and edges (1-dimension cells) bound by a couple of nodes each. In this case, *adjacency matrix* a_{ij}, by definition, is a binary table contained 0 and 1 values, where a unit value $a_{ij} = 1$ indicates that the element i is a neighbour to the element j (they have a common boundary) and $a_{ij} = 0$ otherwise. Importantly, the adjacency matrices are square as they encode correspondence between the same set of elements (same rank or dimensions – between all nodes or all edges in the graph).

Incidence matrix often referred to also as a *boundary operator* $\hat{\partial}$ encodes the relations between the elements of various ranks or dimensions. In the graph (1-complex Kozlov (2008)) ∂_{ij} contains node numbers as its rows i and edge numbers as its columns j. Any non-zero element in a row $\partial_{ij} \neq 0$ indicates that the edge j has node i at its boundary (at one of its ends). Another difference between the adjacency and incidence matrices is that the latter should also express *orientations* of cells Grady and Polimeni (2010) assigned as negative values (usually -1) for some of their values. Fig.6 shows, as an example, a simple graph with its adjacency and incidence matrices. The following relation connects the adjacency and incidence matrices:

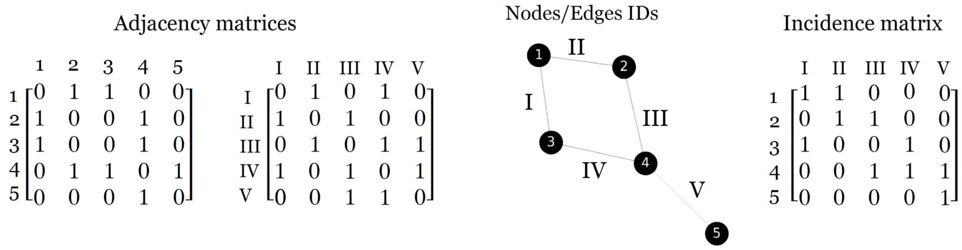

Figure 6: Example of a simple graph with its adjacency and incidence matrices.

$$a_{ij} = \partial_{ij}^T \cdot \partial_{ij} - d_{ij}, \tag{1}$$

where $\hat{d}$ is a diagonal degree matrix containing the number of neighbours for each element d_{ii}, and $d_{ij} = 0$, for any couple of indices $i \neq j$. Hence, the incidence matrices ∂_{ij} solely determine the structure of a simple graph, as each element of a degree matrix d_{ii} can be obtained by finding non-zero elements in the column i of ∂_{ij} and then defining incident elements j in the rows corresponding to each boundary. The conjugate value

$$L_{ij} = \partial_{ij} \cdot \partial_{ij}^{T} \tag{2}$$

is called a *discrete* or *combinatorial Laplacian* of a graph and plays an essential role in graph theory. In the theories of electrical networks, L is also known as Kirchhoff's matrix expressing the number of spanning trees Bollobas (1998) in a network. The Laplacian can also be expressed via the degree and adjacency matrices as

$$L_{ij} = d_{ij} - a_{ij} \tag{3}$$

As combinatorial complexes Kozlov (2008) are the extension of graphs to higher-dimensional structures, they can be represented as a set of adjacency and incidence matrices that will be discussed in Section 2.

1.3. Computational representation

It can be seen even from Fig. 1 that for an average graph, there are numerous 0 values in its adjacency and incidence matrices far outnumbering non-zero elements. Hence, most of the information contained in the full matrices is excessive, and it can be encoded in a more compact form of *sparse* matrices. There are several common approaches to the creation of the sparse matrices, but, importantly, all of them are nothing more than computationally efficient representations of the $\hat{\partial}$ and $\hat{d}$ matrices. In particular, the first computation PCC standard (*pcc1s*) (can be used in the CPD code) uses the format $\{i,\ j,\ value \neq 0\}$ for each non-zero element in the adjacency and incidence matrices.

2. Polytopal Cell Complexes and Discrete Combinatorial Spaces

In 2020-2025 years Jivkov et al. (2023); Borodin et al. (2024d); Boom et al. (2022); Berbatov (2023); Borodin and Jivkov (2019), the application of tools from discrete exterior calculus (DEC) Hirani (2003); Yavari and Goriely (2012); Berbatov et al. (2022), and the entire discrete framework for analysis of processes taking place on the PCC elements have been significantly developed, allowing for direct applications in the mechanics Boom et al. (2022); Dassios et al. (2018); Jivkov and Yates (2012) and physics Berbatov et al. (2022); Jivkov et al. (2023); Šeruga et al. (2020) of solids.

2.1. From tessellation of space to a polytopal cell complex

A material microstructure can be mathematically represented as a PCC possessing the corresponding combinatorial and topological characteristics Kozlov (2008). In a very general view, it is considered a collection of connected elements of different dimensions: 3D *grains*, 2D *interfaces* (including grain boundaries (GBs)), 1D *triple junctions* (TJs) Priester (2013), and 0D *quadruple points* (QPs) Frary and Schuh (2005). A digital model of a microstructure can be created by tessellating a given spatial domain – 2D cut or 3D volume – into convex polytopes so that the shapes and sizes of these polytopes statistically represent the elements of a real microstructure using software such as Neper Quey (2021). A tessellation $\mathcal{T}$ does not formed a PCC $\mathcal{M}$, as a mathematical object, by itself, containing *polyhedrons* (3D polytopes) representing grains, *faces*, (2D polytopes) representing GBs, *edges* (1D polytopes) representing TJs, and *nodes* (0D polytopes) representing QPs. PCC is a further algebraic abstraction of a tessellation, being a discrete material model. Fig. 7 shows an example of the possible variety of PCCs representing complex composite material microstructures.

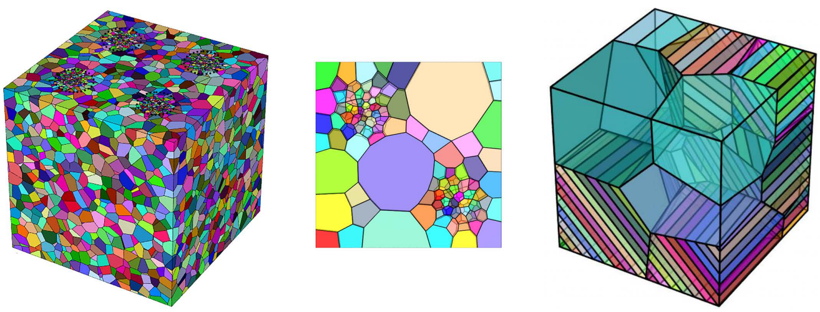

Figure 7: A sketch of various PCC tessellations $\mathcal{T}^3$.

In the language of algebraic topology, the polytopes of dimension p, $p \in \{0, 1, 2, 3\}$ are referred to as *p-cells* σ_p and $\mathcal{M}$ is referred to as a *polytopal* *cell complex* (PCC) Kozlov (2008); Fritsch and Piccinini (1990). An assembly of polyhedra, denoted by $\mathcal{M}$, is a geometric realisation of a combinatorial structure referred to as a polytopal cell complex in algebraic topology May (1999); Kozlov (2008). From a topological perspective, a microstructure is a 3-complex $\mathcal{M}_3$ – a collection of cells with different topological dimensions. A p-cell in $\mathcal{M}$ represents a vertex, an edge, a polygonal face, and a polyhedron for $p = 0, 1, 2$ and 3, respectively. These are illustrated in Fig. 8. Let σ_p

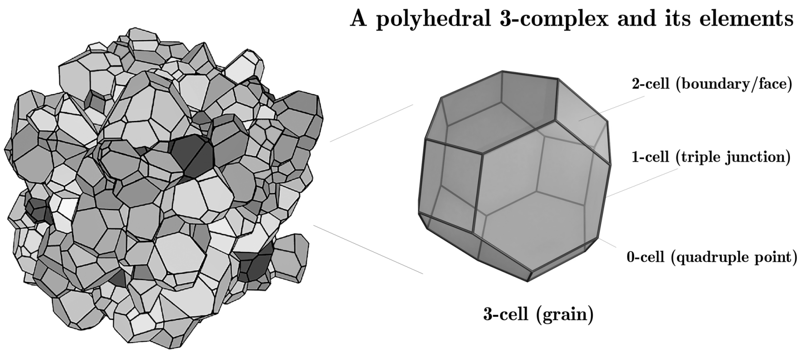

Figure 8: A sketch of a 3-complex with its p-cells of different dimensions Borodin and Jivkov (2019).

denote an arbitrary p-cell and N_p be the number of p-cells in $\mathcal{M}$. The relation $\sigma_p \prec \sigma_{p+1}$ means that σ_p is incident to (on the boundary of) σ_{p+1}. A formal linear combination of p-cells is called a p-chain and is denoted by c_p. The p-chains in $\mathcal{M}$ form a vector space, denoted by C_p, with individual p-cells as a basis. The cells of the complex are assigned orientations. A standard way is to decide on a consistent orientation of all top-dimensional cells, e.g., to select the positive orientation to be from the interior to the exterior of the 3-cells and assign arbitrary orientations for all lower-dimensional cells Cooke and Finney (1967). There are exactly three options for the relation between σ_p and σ_{p+1} in an oriented complex: $\sigma_p \not\prec \sigma_{p+1}$, encoded by 0; $\sigma_p \prec \sigma_{p+1}$ and they have consistent orientations, encoded by 1; $\sigma_p \prec \sigma_{p+1}$ and they have opposite orientations, encoded by -1. These relations are collected in a set of boundary operators, ∂_p for $p \in \{1, 2, 3\}$, which describe simultaneously the topological structure of $\mathcal{M}$ and operations on chains: ∂_p maps p-chains to $(p-1)$-chains, $\partial_p : C_p \to C_{p-1}$, with the property $\partial_{p-1} \circ \partial_p = 0$, i.e., the boundary of a boundary is empty. Algebraically, ∂_p is an incidence matrix of dimensions $N_{p-1} \times N_p$. Furthermore, $\partial_0 = 0$, i.e., the boundaries of 0-cells are empty, and $\partial_4 = 0$, i.e., the 3-cells are not boundaries of higher-dimensional cells.

Fig.9 shows a large tessellation of a cylinder containing 100,000 3-cells (grains), 780,707 2-cells (faces), 1,361,553 1-cells (edges) and 680,847 0-cells (nodes). It can be seen that the number of elements of all PCC dimensions $p \in \{0, 1, 2, 3\}$ can 10-20 times overcome the number of its 3-cells (volumetric elements).

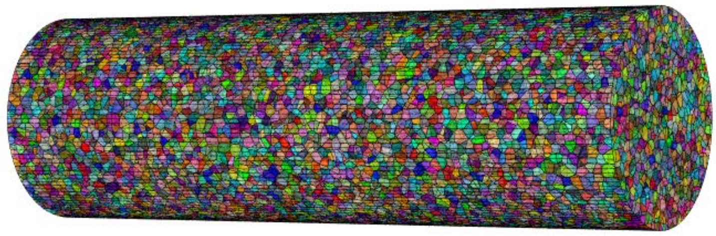

Figure 9: Voronoi tessellation $\mathcal{T}^3$ of a cylinder containing 100,000 3-cells (volumes or grains), 780,707 2-cells (faces), 1,361,553 1-cells (edges) and 680,847 0-cells (nodes) constructed by Neper software Quey (2021).

2.2. Subcomplexes

2.2.1. Local clusters of cells

Each 3-cell with all its incident cells of lower dimensions can be considered a separate PCC or *subcomplex* of a parent PCC. The generalisation of this idea is the creation of local grain *clusters* $\mu^{cl} \in \mathcal{M}$ that consist of several neighbouring grains. By definition, the k-neighbours of a given grain are the surrounding grains which have a common grain boundary with the grain's $(k-1)$-neighbours but do not have any common GBs with the grain's neighbours of lower orders. It can be shown that in the experimentally observed structures, the k-neighbours do not necessarily have common boundaries with $(k+1)$-neighbours or other k-neighbours (i.e. see Fig. 10). While such cases of 'separated' grains are rare and statistically negligible, they demonstrate the topological complexity of real microstructures. With this definition, a grain is its own 0-neighbour, and its 1-neighbours are the grains sharing a boundary with it. Clearly, each grain with the cluster of its k-neighbours can be considered separately as a *reduced PCC* or a *p-subcomplex* $\mu_k \in \mathcal{M}^3$ of a given PCC defined by the smaller sets of σ_p cells and the corresponding reduced sets of incidence matrices ∂_p (boundary operators).

2.2.2. Subcomplexes generated by geometric cuts

Another type of subcomplexes μ^{cut} can be induced by the geometric intersections of the initial tessellation $\mathcal{T}$ with geometric lines, planes and volumes.

Plane Cut Subcomplexes

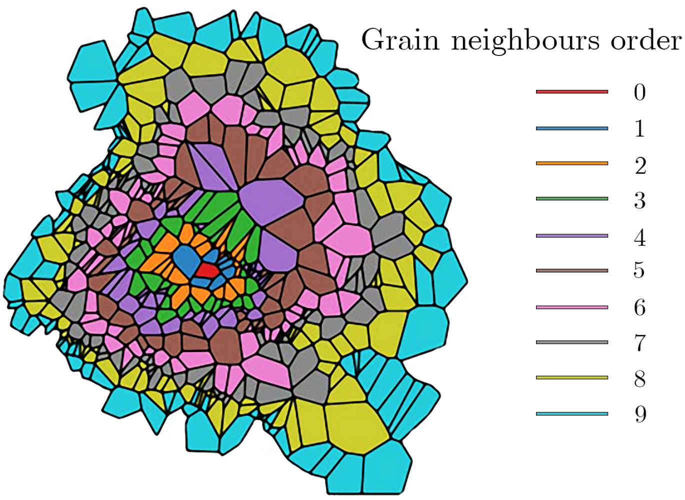

Figure 10: A sketch of grain neighbours up to 9th order in a real-like microstructure of a polycrystalline material Bushuev et al. (2025).

In particular, the modelling of macroscopic cracks Borodin et al. (2025) requires the introduction of PCC-wide crack planes into PCC. This can be achieved by the four-stage procedure shown in Fig.11: (i) Creation of a PCC $\mathcal{T}^p$ as a geometric realisation of a p-complex $\mathcal{M}^p$ embedded in a p-dimension continuous space $\mathcal{R}^p$; (ii) Extraction of a subcomplex of this PCC covering a plane associated with the macrocrack geometric plane $\{a, b, c, D\}$; (iii) Creation of a lower-dimensional PCC $\mathcal{M}^{p-1}$ formed by traces of the plane subcomplex cells and inherited the colouring and labelling (defect microstructure) of the initial p-complex $\mathcal{M}^p$; (iv) Introducing a half-plane with an edge associated with the macrocrack tip (Fig.11). By definition, the macrocrack tip here is associated with the only edge of the plane subcomplex which is not contained in one of the boundaries of the initial tessellation volume (a PCC's geometric boundary). The shortest geometric distance between this tip edge and the opposite subcomplex edge is equal to the macrocrack length a_l, and the internal area denoted as S_a formed by the three intersections of the half-plane cut with the PCC geometric boundaries and the tip-edge.

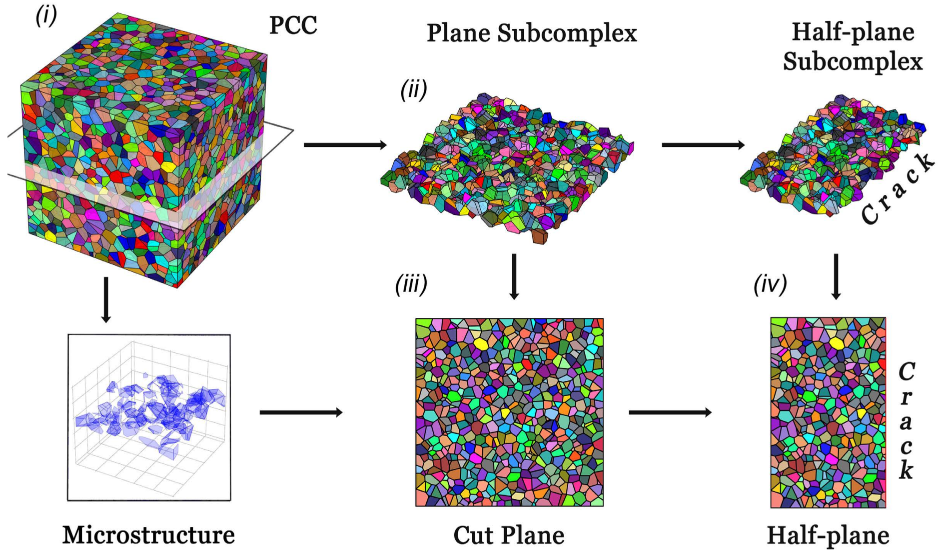

Figure 11: Obtaining a plain section Borodin et al. (2024d) of a cube PCC with the inheritance of its initial colouring: (i) Space tessellation $\mathcal{T}^3$ of a cube with 3,000 Voronoi polyhedrons represented a PCC as a geometric realisation of a corresponding 3-complex $\mathcal{M}^3$; (ii) Subcomplex of this PCC covering a geometric plane $\{a, b, c, D\}$; (iii) A plane PCC cut as a new corresponding $(p-1)$-PCC $\mathcal{T}^2$ created by traces of the p-cells of the initial PCC $\mathcal{T}^3$; (iv) Half-plane cut as a $(p-1)$-PCC with a special edge associated with the macrocrack tip.

3. Labelling and Special Cell Types

A constructed polytopal cell complex (PCC) $\mathcal{T}^p$ or its algebraic representation $\mathcal{M}^p$ Kozlov (2008); Jivkov and Yates (2012); Boom et al. (2022) is a discrete combinatorial space where physical processes may occur. Its topology and, once embedded in a metric space as $\mathcal{R}^p$, geometry, impose specific restrictions on the physical processes but do not determine them in any sense. The basic idea of the PCC Processing Design (*aka* **C**ombinatorial **P**rocessing **D**esign) or *CPD code* is the addition of graphs of special cells, representing material microstructure, to the p-cell skeletons of various dimensions of a PCC given as a combinatorial discrete space.

3.1. Skeletons, their state vectors and special cell types

The collection of all q-cells, $0 \leq q \leq p$, together with the lower-dimensional cells on their boundaries is referred to as the *q-skeleton* of a PCC $\mathcal{M}^p$, denoted by $\mathcal{M}_q$: $\mathcal{M}_3$ coincides with $\mathcal{M}^3$; $\mathcal{M}_2$ is formed by all 2-cells, 1-cells, and 0-cells of $\mathcal{M}^3$; $\mathcal{M}_1$ is formed by all 1-cells and 0-cells of $\mathcal{M}^3$; and $\mathcal{M}_0$ is formed by all 0-cells of $\mathcal{M}^3$.

A sub-structure of a polycrystalline assembly, like the one ($\mathcal{M}_3$) shown in Fig. 9, with elements of dimension q is a geometric realisation of $\mathcal{M}_p$. Consider a criterion by which the elements of the p-substructure of a given polycrystalline assembly are classified into sets with distinct characters. Each set corresponds to a subset of $\mathcal{M}_p$, and the union of all sets is $\mathcal{M}^p$. Each set of elements can be described by boundary operators $\hat{\partial}_p$ for $p \in \{1, 2, 3\}$ that account for the connectivity and the relative orientations only of the elements belonging to the set. The neighbourhood of p-cells can be algebraically represented by adjacency matrices $\hat{a}_p$ Bollobas (1998); Durrett (2010). They are related to the matrices of the boundary operators $\hat{\partial}_p$ via the diagonal matrix of the corresponding p-cell degrees $\hat{d}_p$ as

$$\hat{a}_p = \hat{\partial}_p^T \cdot \hat{\partial}_p - \hat{d}_p, \tag{4}$$

where $\hat{d}_p$ is a square matrix of the same size as $\hat{a}_p$ contained the corresponding numbers N_p of p-cell neighbours as its diagonal elements, and zeroes on the all non-diagonal elements, and $\hat{\partial}_p^T$ is the transposed incidence matrix $\hat{\partial}_p$. A 2-skeleton that represented all material interfaces is often of particular interest for the microstructural analysis. Skeletons of all dimensions are illustrated in Fig.3, on the example of a PCC represented by a single polyhedron.

3.1.1. Assigned special cells

The simplest way to introduce a structure represented by a network or graph is a simple labelling of the PCC cells by assigning non-zero values to the corresponding state vectors S_p^a as described in Section 1. Here, the index a denotes the *assigned* type of labelling in contrast with the *generated* indexed by g and *induced* indexed by i, which will be discussed further. We will use the terms labels and colours interchangeably.

Initially, all the PCC skeletons are supplied with zero state vectors $S_p^a = \{0, 0, ..., 0\}$ of sizes corresponding to the number N_p of p-cells in the corresponding PCC $\mathcal{M}^p$. Then, during the *labelling* process referred to as a 'PCC processing', some of the elements of S_p^a change their values with IDs greater than zero. For instance, interfaces of a polycrystalline material or microcracks can be classified into several subsets labelled by different IDs, with respect to their physical energies or GB disorientations. In binary cases, such as low-angle and high-angle grain boundaries (LAGBs and HAGBs) explored in Ref. Borodin and Jivkov (2019); Zhu et al. (2021), the corresponding state vectors in the CPD code contained only $\{0, 1\}$ values which can be assigned either randomly (i.e. see the random generation function in the PCC_Processing module of the CPD code) or according to some extremal principle based on the combinatorial measures, which are discussed below. Fig. 12 show a sketches of a cubic 2-skeleton $\mathcal{M}_2$ of $\mathcal{M}^3$ embedded as $\mathcal{R}^2$ a 3-dimension space; its representation as a *lattice* by assigning nodes to the barycentres of faces (2-cells) and edges (1-cells) connecting threes and fours of 2-cells possessing common junction; and finally a graph of *special* 2-cells assigned by the corresponding non-zero labels in the state vector S_2^a. In practice, the assignment of a fraction of f non-zero elements to any S_p^a

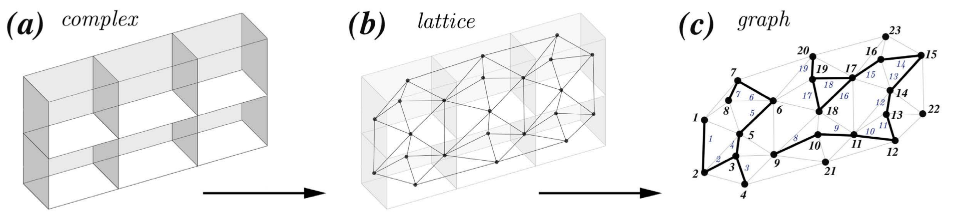

Figure 12: A sketch of (a) a 2-skeleton of a small cubic PCC; (b) its representation as a lattice by assigning nodes to the barycentres of cells and edges connecting all cells possessing common lone junction; (c) a graph of special 2-cells assigned by the corresponding state vector S_2^a Borodin and Jivkov (2019).

($p \in \{0, 1, 2, 3, \}$) can be implemented by a simple random generation of $f \cdot N_p$ non-zero values and replacement by them of the corresponding elements in the S_p^a state vector. Alternatively, the assignment of non-zero values to the

S_p^a elements can follow specific rules dependent on local defect patterns in the PCC (p-cell indexes Borodin et al. (2021)) or global network topology and combinatorial measures Borodin and Jivkov (2019); Zhu et al. (2021) which will be discussed below in Subsection 4.

Chains of assigned special cells and their size distribution

In the works Borodin et al. (2024d, 2021), only the 'inclusion flakes' covered strictly one face each (separating independent special 2-cells in $\mathcal{M}^3$) have been considered. However, in most practical cases, the inclusions have a geometry of narrow strips with a log-normal length distribution $\nu(l)$. Similarly, the chains of 1-cells or 3-cells with their length distributions can be considered.

To represent the chains of connected special p-cells with a given lengths distribution $\nu(l^p)$ (expressed in the number of p-cells) the following procedure can be applied: (i) Split the total length of strips in a 'powder' (expressed in the number of p-cells) over the number of separate bins N_b^p. It can be done by (ii) assigning for each of these bins the fraction f_p of special p-cells according to the given distribution of σ_p numbers in each bin $\nu^p(n)$ possessed the given values of its mean μ_p and dispersion σ_p, and then (iii) division of the obtained number by l^p equal to the index number of this bin as $\nu(l^p) = \nu(n^p)/l^p$.

In the case of the log-normal chain length distribution:

$$\nu(l_k^p) = \frac{1}{n_k^p \cdot l_k^p \cdot \sigma_p\sqrt{2\pi}} \cdot \exp\left(-\frac{(\log n_k^p - \mu_p)^2}{2\sigma_p^2}\right). \tag{5}$$

Here $\nu(l^p)$ is the frequency of appearance in the 'powder' of chains of σ_p with a particular length l_k^p (expressed in the number of p-cells $\#\sigma_p = k$), and n_k^p is the total number of special p-cells in each bin.

Computationally, the random placement of chains of special p-cells in the corresponding p-skeletons $\mathcal{M}_p$ of a PCC $\mathcal{M}^p$ can be implemented by a random walker (RW) algorithm Klafter and Sokolov (2011), where the initial p-cell for each new chain is arbitrarily chosen among the whole set of p-cells in the PCC. Then, using the adjacency $\hat{a}_p$ and incidence $\hat{\partial}_p$ matrices of the PCC, the algorithm Borodin (2025) at each step looks for all the p-cell's neighbouring cells and chooses randomly a new p-cell among them assigning its type as 'special' (non-zero) in the corresponding state vector S_p^a. This procedure repeats before the number of newly assigned consequent special p-cells reaches the chain length l_k^p expressed in the number of p-cells and is taken from the given chain distribution $\nu(l_k^p)$, such as a log-normal one (Eq.(5)). Then the whole process of random assignment of an initial special face and the RW procedure repeats before exhausting all chains in a powder, in accordance with its size distribution. Ultimately, the information about all special p-cells associated with the chains of microstructural elements (such as inclusions σ_2 in Borodin et al. (2025)) is encoded in the corresponding S_p^a vectors. For each newly chosen p-cell number, the corresponding element of S_p^a is replaced with $'1'$, indicating the presence of a special microstructural component.

In some cases, during the RW procedures, several 'special' types can be assigned to the same p-cells because of the intersection of chains, indicating the appearance of an *agglomeration* in this p-cells with a *power* w_a^p equal to the number of special microstructural components assigned to this cell by the RW algorithm.

For PCC subcomplexes $\mu_i^p \in \mathcal{M}^p$, the microstructure and the corresponding state vectors $S_p(k)$, where k is a subcomplex ID, can be obtained by the procedure of *microstructure inheritance* from a parent PCC by the covering subcomplexes possessing non-zero state vectors S_2^a. In the case of k-order neighbours (see Subsection 2.2.1) it gives the local combinatorial measures discussed in the following subsection; for PCC cuts, including half-planes (see Fig. 11), it allows estimation of the number of p-cell chains n_t traced the lower-dimension PCC cut (see Subsection 2.2.2) and their average length $< l_t >$.

In the especially important case of 2-cell chains representing the elongated inclusion strips Borodin et al. (2025), agglomerations coincide with the usual agglomerations of inclusions, well-known for the composite materials. Considering a macroscopic crack growth Borodin et al. (2025), the network of elongated inclusions should significantly contribute to a composite fracture toughness K_{IC} via the bridging effect Weibel et al. (2020). The estimation of the average length of inclusions crossing the crack plane $< l_t >$ allows one to determine the additional energies associated with the pulling out of inclusions during the crack opening process Borodin et al. (2025).

3.1.2. Generated special cells

A *generated* cell type represents special material microstructures, e.g., microcracks Borodin et al. (2024d), that depend on the primal assigned structures, e.g., inclusions Borodin et al. (2024d), of special cells encoded in the state vectors S_p^a. Both special and generated sets of p-cells ($p = \{0, 1, 2, 3\}$) in a PCC can be represented as graphs placed on the corresponding p-skeletons $\mathcal{M}_p$ Borodin et al. (2024d, 2021); Kozlov (2008). Similarly to the assigned cell type, any arrangement of generated p-cells can be encoded in a DNA-like Mitchell (2009) generated state vector S_p^g Borodin et al. (2024d, 2021). Fig. 13 from the MATERiA project Borodin et al. (2024d) demonstrates on the examples of 2-PCC how diverse generated cell structures (blue edges) can be if they are obtained by the same rules from different initial structures of assigned special cells (grey edges). A generation or kinetic processes leading to

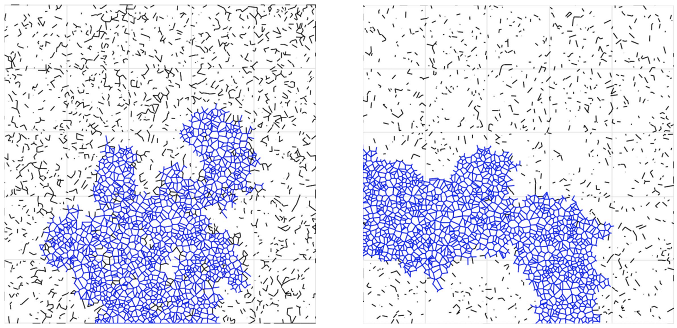

Figure 13: Initially assigned special 1-cells (grey) and 1-cells of the generated type created during a kinetic process (blue). Both structures were obtained by CPD code using the same 2-PCC and the same generation rules.

the appearance of the generated cell types can be very complex and typically is determined by a physical model (set of equations) and some *generation rules*. The generation rules relate physical quantities associated with each p-cell, such as energies, to their generated type in full analogy with the thermodynamic state variables and the equation of states:

$$S_p^g = \text{function(Stress,Temperature,time,..)}$$

. Fig. 14 shows an example of a structure of special 1-cells obtained similarly (the same 2-complex and similar initial microstructure of assigned grey cells) to the structures presented in Fig. 13 but with different generation rules accounting for local stress concentrators Borodin et al. (2024d). The corresponding indexing rules allow the implementation of 'induced' classification of low-dimensional p-cells based on the higher-dimension cell types (Subsection 3.1.3), and, conversely, 'indexing' of higher-dimension p-cells by low-dimension cell types on their boundaries (Subsection 3.2).

3.1.3. Induced classification of low-dimensional cells

In a PCC created by Voronoi tessellation, each 1-cell (edge) is a meet of maximum three 2-cells (faces), and each 0-cell (node) is a meet of maximum four 1-cells (TJs) and six 2-cells (faces)Borodin et al. (2021); Zhu et al. (2021). Such that even a binary $\{0, 1\}$ classification of 3-cells, e.g., as ordinary and special assigned or generated types, induces three different types of adjacent 2-cells: $\{0, 0\}$, $\{0, 1\} = \{1, 0\}$, and $\{1, 1\}$. Similarly, a binary classification of 2-cells $\{0, 1\}$ induces four types of 1-cells J_ω, as meets of $\omega \in \{0, 1, 2, 3\}$ special 2-cells Borodin et al. (2021); Zhu et al. (2021), and 11 different types of 0-cells, as it was shown in Ref. Frary and Schuh (2005). Fig. 15 shows all

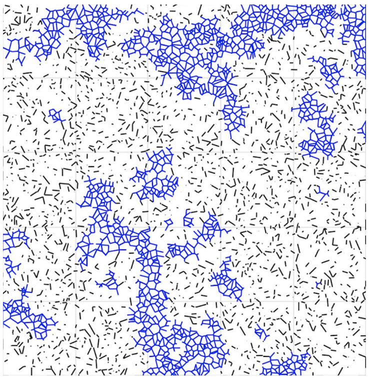

Figure 14: Initially assigned special 1-cells (grey) and 1-cells of the generated type created during a kinetic process (blue). The structure was obtained by CPD code using the same 2-PCC as structures shown in Fig. 13 and similar initial microstructure but different generation rules accounted for local stress concentrators Borodin et al. (2024d).

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possible types of 1-cells in a Voronoi $\mathcal{T}^3$ induced by a binary classification of its 2-cells. Topologically, a 1-cell of J_1 type is an end of the chain of special

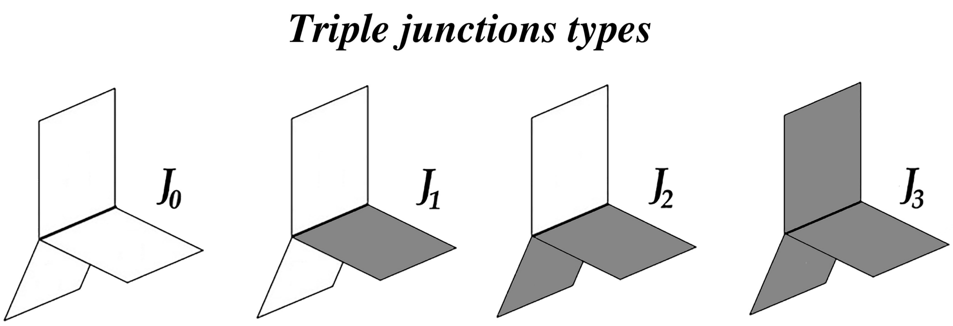

Figure 15: Induced classifications of 1-cell types in a Voronoi 3-PCC Borodin et al. (2021).

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2-cells; J_2 is an 'internal' 1-cell in the network of special 2-cells without branching; and J_3 is an internal 1-cell where the network of special 2-cells branches. The 1-cell types induced by assigned J^a_ω (induced by S^a_p), where assigned 2-cells meet, and generated J^g_ω (induced by S^g_p) where generated 2-cells meet can be distinguish. In such a classification, both assigned and generated networks of p-cells are considered fully independent from each other. Fig. 16

Similarly to the assigned and generated p-cell types, any induced substructure can be encoded in the corresponding S^i_ω state vectors, where the elements are equal to the $\omega \in \{0, 1, 2, 3, ..\}$ value of the corresponding induced type. In

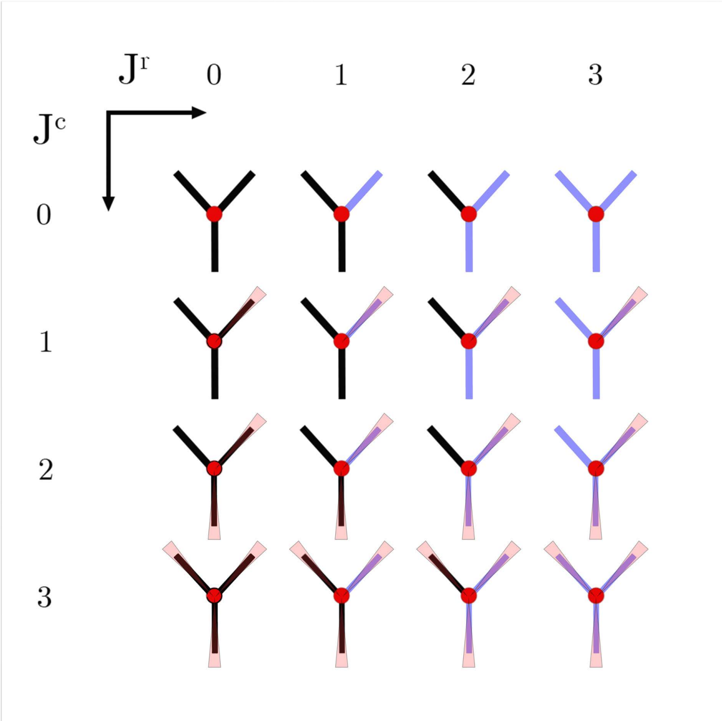

Figure 16: 0-cell types induced by assigned and generated classification of 1-cells Borodin et al. (2024d).

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the case of 1-cells with types induced by the considered binary classification of 2-cells, each S_1^i element is equal to one of the four $\omega \in \{0, 1, 2, 3\}$ numbers corresponding either to its J_ω^a or J_ω^g TJ types. It is clear that to represent the 1-cell structures induced by both assigned and generated (e.g. inclusions and microcracks in composites Borodin et al. (2024d)) cell types, two different $(S_1^i)_a$ and $(S_1^i)_g$ vectors are needed.

It is important to note that the introduced induced cell J-types express not only the structures of special cells (their graphs), but also elements of $'0'$-types associated with the 'ordinary' p-cells and their junctions. For representation solely a graph structure of special p-cells without the PCC 'space' of ordinary cells surrounding them, the other *derivative* induced D-types should be introduced Borodin et al. (2024d).

Consider only one 2-cells network of a special types, J_0, J_1 and J_2 such that this sub-structure can be classified according to the number of incident 2-cells only of a special type. Thus, the 'free leafs' have type D_1, branches or loops connected by J_2 junctions have type D_2, and J_3 junctions have type D_3. Fig. 17 sketches all possible D-types of the 1-cell junctions of 2-cells in a 3-PCC based on the Voronoi tessellation $\mathcal{T}^3$. Similarly to the induced

Figure 17: A sketch of a 'flake' (coloured pure), an agglomeration (coloured red) with three intersecting special 2-cell chains and various possible D_i types of 1-cells.

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structure of J_ω^p cells, the derivative classifications D_ω^p can be extended to any order of p-cells in a PCC.

3.2. Local indexing of cells

In addition to the types induced by higher-dimension p-cells on their incident lower-dimension cells (boundaries), any integer values associated with p-cells can *index* both higher-dimension and lower-dimension cells incident to them. A particular example described in Borodin et al. (2021) is the B_L and Q_L triple junction (1-cell) indices assigning numbers both for grain boundaries (2-cells) and quadruple points (0-cells). Physically, if special 1-cells are associated with the local concentration of inclusions or cracks adjacent to a TJ, their indices represent local stress concentrators around interfaces, increasing the propensity for cracking and formation of pores.

Let ζ_1, ζ_2,.., ζ_τ denote the numbers of $(p-1)$-cells on the boundary of a p-cell of types $\{1, 2, 3, ..\tau\}$, respectively. The $(p-1)$-index characterising the concentration of special cells around a single p-cell is defined as

$$B_p = \sum_{k=1}^{\tau} \alpha_k \zeta_k \tag{6}$$

where, it is often convenient to assumed $\alpha_k = k$ – higher incident number of inclusions lead to higher index values Borodin et al. (2021, 2024d). Fig. 18 schematically shows the numbers J of triple junctions (0-cells) of 1-cells of different types J_k^a, $k \in \{0, 1, 2, 3\}$ alongside the $B = B_0$ indices and configuration entropies S (which will be discussed in Subsection 4.2) induced by 1-cells (edges) of the hexagonal small 2-PCC with a three tailored configuration of the assigned special cells Borodin et al. (2021). It can be written similarly for the indices B_p^a and B_p^g based on the assigned and generated special cell types, respectively. An illustration of GBs (2-cells) with the corresponding TJ (1-cells) 1-indices is shown in Fig.19.

Similarly, if $\eta_1, \eta_2, .., \eta_\epsilon$ denote the numbers of $(p+1)$-cells on the coboundary of a p-cell of types $\{1, 2, 3, ..\epsilon\}$, respectively. The $(p+1)$-index characterising the concentration of special cells around a single p-cell is defined

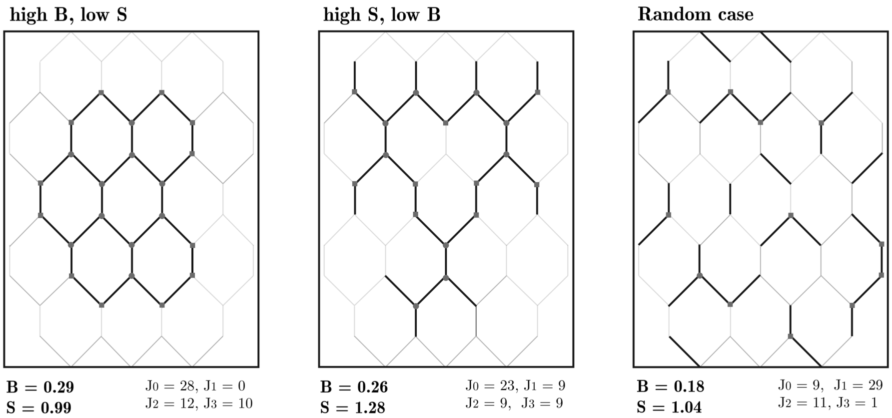

Figure 18: A sketch of a hexagonal small 2-PCC with a three tailored configuration of the assigned special cells Borodin et al. (2021). Shown are the values of $B = B_0$ indices and configuration entropies S (which will be discussed in Subsection 4.2) corresponding to each of the microstructures, including the random one (right). Also shown J numbers of 0-cells of different types J_k^a, $k \in \{0, 1, 2, 3\}$ induced by the assigned special 1-cells.

as

$$Q_p = \sum_{k=1}^{\epsilon} \beta_k \eta_k, \tag{7}$$

where, similarly to B_p index, it is often convenient to assumed $\beta_k = k$ – higher incident number of inclusions lead to higher index values Borodin et al. (2021). These definitions do not account for the variety of GB geometries, which may range from triangles to polygons with many edges. It is therefore appropriate to normalise them using the number of neighbours of the p-cell, W_b^p, which corresponds to its topological weight or simply the total number

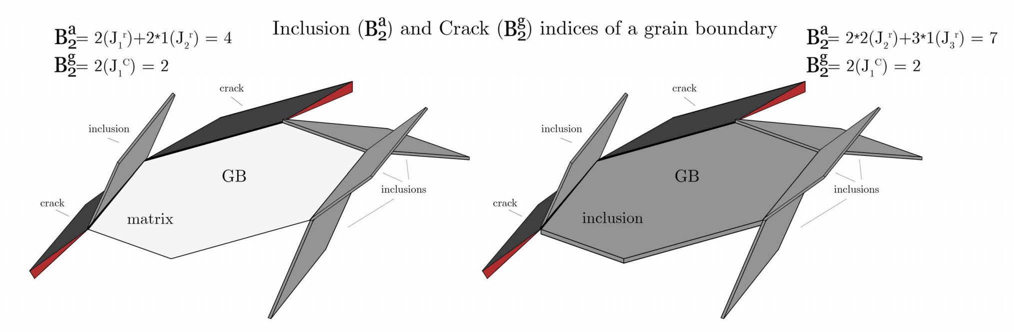

Figure 19: A sketch of a grain boundary does not possess an assigned special cell type (left) and a grain boundary possessed the assigned cell type (right) with the corresponding values of TJ indices B_2^a and B_2^g Borodin et al. (2024d).

of $(p-1)$-cells on its boundary introduced in Forman (2003). The normalised indices $\widetilde{B}_p = B_p/W_p$ and $\widetilde{Q}_p = Q_p/W_p$ have values less than or equal to 1, which makes them more convenient for computations.

4. Combinatorial Measures and Topology of Cells

4.1. Fractions of special cells

The most valuable statistical parameter related to the state vectors S_p^a,S_p^g and, simultaneously, the simplest statistical measure is the *fraction* of non-zero elements in the state vectors representing special p-cell types. The fractions are dimensionless and express the numbers of special p-cells of different types in the corresponding PCC. Let us use notation η^p for the fraction of assigned special p-cells – non-zero elements in S_p^a and f^g for the fraction of the generated special p-cells – non-zero elements in S_p^g:

$$\eta^p = \frac{N_a^p}{N^p}, \quad f^p = \frac{N_g^p}{N^p}, \tag{8}$$

where N_a^p is the number of p-cells of the assigned special type F_ω^a, N_g^p is the number of p-cells of type F_ω^g, and N_p is the total number of p-cells in the PCC. Additional notations, η_α^p and f_α^p, can be used for the fraction of p-cells containing agglomerations Borodin et al. (2025) of assigned and generated special cells, respectively.

Often, the relations between the combinatorial η^p, f^p fractions and the corresponding physical variables, such as volume fraction of inclusions Borodin et al. (2021), can be established. Moreover, the fractions of special cells can be written as functions of time Borodin et al. (2024b) or physical characteristics defined the process, such as accumulated plastic strain Borodin and Jivkov (2019); Zhu et al. (2021). Such a functional maps a purely combinatorial process that happened in the discrete PCC space to real macrostructural changes, making possible comparison of obtained results with experimental data Zhu et al. (2021, 2023a, 2024).

More non-trivial are the fractions corresponding to the induced state vectors $(S_1^i)_a$ and $(S_1^i)_g$ (see Subsection 3.1.3) obtained as derivatives from the S_p^a and S_p^g using the incidence matrices $\hat{\partial}_p$ of the PCC. The fractions of the induced p-cells of the corresponding type J_ω^a or J_ω^g, $\omega \in \{0, 1, 2, 3\}$ is

$$j_\omega^p = \frac{N_\omega^p}{N^p}, \tag{9}$$

where N_ω^p is the total number of J_ω^a or J_ω^g types equal to the number of the $\omega \in \{0, 1, 2, 3\}$ values in the corresponding $(S_1^i)_a$ and $(S_1^i)_g$ vectors; N^p is the total number of the corresponding p-cells in the PCC.

The induced j_ω^p fractions are usually much more informative compared with the 'trivial' fractions η^p and f^p. Their changes during the structure evolution process can reveal unanticipated features related to the material or its particular processing type Borodin and Jivkov (2019); Zhu et al. (2021, 2023a).

4.1.1. Configuration space

In addition to the induced j^p_ω, $\omega \in \{0, 1, 2, 3\}$ fractions, a derivative fractions d^p_k, $k \in \{1, 2, 3\}$ can be introduced representing the derivative induced D-types discussed in Subsection 3.1.3. The fractions d^a_k and d^g_k induced by the assigned and generated p-cell types can be calculated directly from the j^a_ω and j^g_ω fractions as

$$d^a_k = \frac{j^a_k}{1 - j^a_0}, \quad d^g_k = \frac{j^g_k}{1 - j^g_0}. \tag{10}$$

Therefore, at large values of j^p_0, even small fractions j^p_ω can give large fractions of the corresponding d^p_k. The main difference between the d_k classification and the j_ω fractions is that in the former, exclusively special p-cells are considered as if there were no other p-cell types in the PCC.

The (d_1, d_2, d_3) TJ types degree space describes the global topology of an inclusion network. Fig. 20 shows an example of such a network degree space (NDS) representing assigned S^a_2 structures of separate inclusion 'flakes' in comparison with elongated chains of inclusions.

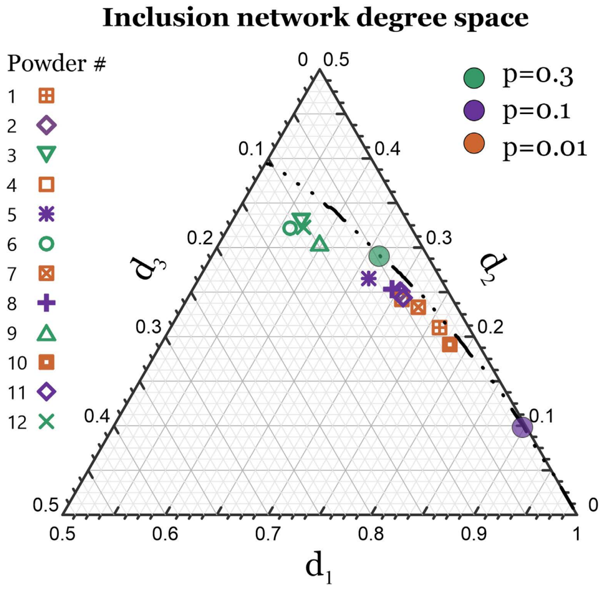

Figure 20: Inclusion S^2_a network topologies for the random distribution of separate inclusion flakes (dash-dot curve) in comparison with the 12 sets $\{\mu, \sigma, p\}_k$ of elongated inclusion strips Borodin et al. (2024d)

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4.2. Configuration and structural entropy

The configurational or structural entropy S_{conf} is a combinatorial measure designed for quantification of diversity or information content encoded in the state vectors S_p, including the induced cell types S_p^i. It is independent of the PCC size and related directly to the system's information content and is formally equivalent to the standard definition of Shannon (information) entropy Shannon (1948, 1997). Specifically, it measures the complexity of the corresponding network of special cells, i.e., the amount of information required to describe it. Higher values of entropy indicate larger complexities. Borodin et al. (2021); Frary and Schuh (2005); Zhu et al. (2021).

$$S_{conf}^{p,q} = -\sum_{\omega=1}^{\omega_{max}} j_\omega^p \log j_\omega^p, \tag{11}$$

where p-cells induce labelling on the q-cells ($q \leq p$) and ω numerates special cell types, and j_ω any fraction of special cells, such that holds $\sum_{\omega=1}^{\omega_{max}} j_\omega = 1$. In particular, for assigned cell types with the fraction η^p:

$$S_{conf}^{p,p} = -\eta^p \log(\eta^p) - (1 - \eta^p) \log(1 - \eta^p), \tag{12}$$

and for TJ types induced by grain boundaries Frary and Schuh (2004); Borodin and Jivkov (2019); Zhu et al. (2021):

$$S_{conf}^{2,1} = -(j_0^1 \cdot log_2 j_0^1 + j_1^1 \cdot log_2 j_1^1 + j_2^1 \cdot log_2 j_2^1 + j_3^1 \cdot log_2 j_3^1), \tag{13}$$

where j_ω^1 are the fractions of the four TJ possible types (see Fig. 15. Since information entropy is typically measured in bits, the base of the logarithm in Eqs. (11)-(13) is 2. Entropy values determined with this base can be converted to any other required base. The entropy S_{conf} is a non-monotonous function of the assigned fraction of special p-cells, with a maximum at $\eta^p = 0.5$ independent of the material's microstructure Borodin and Jivkov (2019). In a multi-dimension structures, such as $\mathcal{M}^2$ or $\mathcal{M}^3$, special cells of maximum dimension can induce J_ω^a or J_ω^g types to $(p-1)$-cells, $(p-2)$-cells,..,0-cells. The S_{conf}^p entropy value increases with each decrease in dimension p, in particular, S_i^0 contains 25% more information compared to S_i^1 Frary and Schuh (2005). Hence, the S_i^0 can be referred to as a *structural* entropy that measures most of the configuration information stored in the discrete structure. Moreover, the entropy can be decomposed into two parts Viana (2008):

$$S_{conf}^p = S_m^p + S_d^p, \tag{14}$$

where S^p_m is a geometric mean part given by

$$S^p_m = \frac{1}{n_p} log_2(j^p_0 j^p_1 j^p_2 \dots j^p_{n_p}), \tag{15}$$

and S^p_d is a divergence part, describing the deviation of special cell fractions from a homogeneous distribution, given by Viana (2008):

$$S^p_d = \frac{1}{n_p} \sum_{k<l} (j^p_k - j^p_l) log_2 \left(\frac{j^p_k}{j^p_l} \right). \tag{16}$$

4.2.1. Reference configurations and deviations from them

For any particular fraction η^p, the configurational entropy can have maximum and minimum values denoted by S^p_{max} and S^p_{min}, respectively. Notably, S^p_{conf} is maximum when the fractions of all cell types are equal (see Eq. (11)). In addition, S^p_{rnd} is the value of the entropy of randomly distributed special p-cells with fraction η^p.

The random distribution of special cells in a statistically representative PCC provides a benchmark for characterising all other distributions of special cells Borodin et al. (2021); Frary and Schuh (2005, 2004). In particular, the fractions of TJs (1-cells) in a Voronoi tessellation as functions of fraction η^2 of special cells in the case of their random distribution satisfy the equations Frary and Schuh (2005):

$$j^1_0 = (1-\eta^p)^3, \quad j^1_1 = 3\eta^p(1-\eta^p)^2, \quad j^1_2 = 3(\eta^p)^2(1-\eta^p), \quad j^1_3 = (\eta^p)^3, \tag{17}$$

where $j_0 + j_1 + j_2 + j_3 = 1$. The entropy of the random distribution can be expressed directly in terms of η^p by substituting Eq. (17) into Eq. (13) Borodin et al. (2021).

Fig. 21 shows the random distributions of the TJs of all types in comparison with the distributions of TJs obtained by maximising S^1_{conf} using the CPD code.

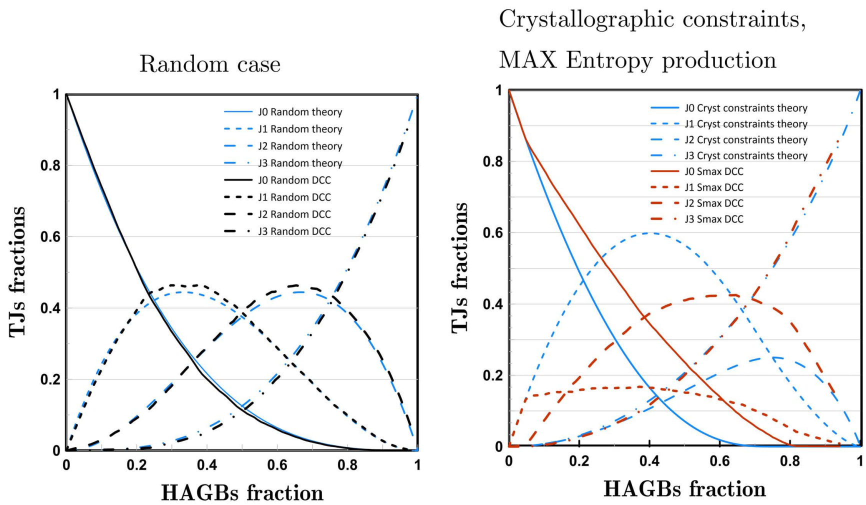


Figure 21: Random distribution of TJ types (left) in comparison with the distribution of TJs crystallographically compatible with possible grain orientations Frary and Schuh (2005) and obtained by maximising $S^{2,1}_{conf}$ in Eq. (13) using CPD code (right).

The divergence S^p_d is negative and measures the distance of a structure from one with maximum configurational entropy. Maximising S^p_d for a given η^p, gives the minimum $S_{min}(\eta^p)$. This is used to construct a reference configuration with minimum configurational entropy by assigning GBs that maximise S^p_d.

4.2.2. Deviation in distributions

It is often essential not only to find the differences between the two distributions reflecting different microstructures for one particular value of η^p or to represent the differences between them graphically, but also to find the quantitative measures for deviations of one distribution from the others. Any fractions of special cells, such as p_i and q_i, can be associated with the discrete probability distributions $P = (p_1, \ldots, p_k)$ and $Q = (q_1, \ldots, q_k)$.

The *Kullback-Leibler divergence* or *relative entropy* is applicable to quantify the *information gain* one will achieve start using the second variable q instead of the first one p:

$$S(p|q) = \sum_i^N p_i \log_2 p_i - \sum_i^N p_i \log_2 q_i = -\sum_i^N p_i \log_2 \frac{q_i}{p_i}. \tag{18}$$

The derivative concept of *mutual information* depends on Kullback-Leibler divergence as

$$I(p;q) = S(p) + S(q) - S(p,q) = S(p) - S(p|q) = S(q) - S(q|p), \qquad (19)$$

which shows the reduction in uncertainty or effective variability of p when q is known, averaging over their joint distribution.

The Hellinger distance Kitsos and Toulias (2017) is another measure alternative to the Kullback-Leibler divergence, which can quantify the similarity between two distributions as

$$H(P,Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^{k} (\sqrt{p_i} - \sqrt{q_i})^2}. \qquad (20)$$

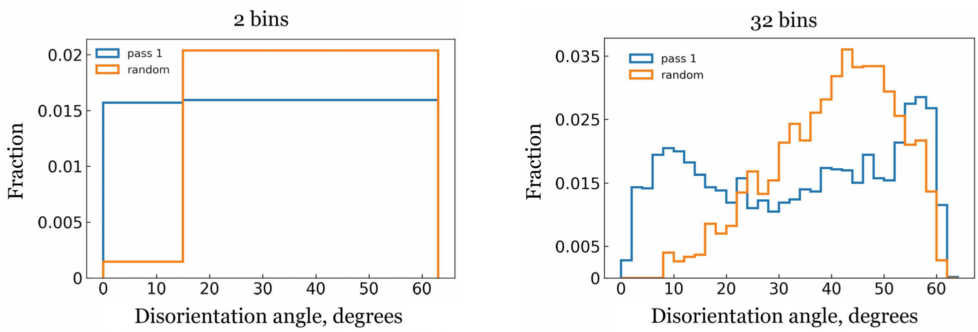
The number of bins representing the P and Q distributions is one of the critical factors affecting the distances. In some cases, such as the binary case of special (with fraction p) and ordinary (with fraction $q = 1 - p$) cells, only two bins are needed, and the Hellinger distance H_2 fully represents the considered distribution.

$$H_2 = \sqrt{p} - \sqrt{q}. \qquad (21)$$

In study Ref.Bushuev et al. (2025) the experimentally observed GB disorientations Priester (2013) were represented as a distribution $P = (p_1, \ldots, p_{32})$ over 32 bins. Its deviation from the theoretical Mackenzie density function Mackenzie (1958) for randomly oriented grains was given as H_{32} Hellinger distance calculated as

$$H_{32} = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^{32} (\sqrt{p_i} - \sqrt{q_i})^2}. \qquad (22)$$

Figure 22 shows the differences between distributions corresponding to the distances H_2 and H_{32} for experimental Bushuev et al. (2025) and the random distribution Mackenzie (1958) of grain orientations.



(a) (b)

Figure 22: Difference between disorientation angle distributions with (a) 2 and (b) 32 bins for experimental data (blue curve) and the random distribution of grain orientations (orange curve) Bushuev et al. (2025).

The H_{32} distance conveys more information about the distribution inside the range of disorientations with high angles.

4.3. Topological analysis of cell structures

The boundary operators $\widehat{\partial}_p$, introduced in Section 2 can be used to form combinatorial Laplacians that map p-chains to p-chains, $\widehat{L}_p : C_p \to C_p$, for $p = \{0, 1, 2, 3\}$. Algebraically, these are matrices of dimensions $N_p \times N_p$ obtained by:

$$\widehat{L}_p = \widehat{\partial}_p^T \circ \widehat{\partial}_p + \widehat{\partial}_{p+1} \circ \widehat{\partial}_{p+1}^T. \tag{23}$$

The Laplacians of $\mathcal{M}_q$ are calculated by Eq. (23) with $\widehat{\partial}_{q+1} = 0$. For example $\mathcal{M}_2$ has $\widehat{L}_0(\mathcal{M}_2) = \widehat{\partial}_0 \circ \widehat{\partial}_0^T \equiv \widehat{L}_0$, $\widehat{L}_1(\mathcal{M}_2) = \widehat{\partial}_1^T \circ \widehat{\partial}_1 + \widehat{\partial}_2 \circ \widehat{\partial}_2^T \equiv \widehat{L}_1$, and $\widehat{L}_2(\mathcal{M}_2) = \widehat{\partial}_2^T \circ \widehat{\partial}_2 \neq \widehat{L}_2$, while $\mathcal{M}_1$ has $\widehat{L}_0(\mathcal{M}_1) = \widehat{\partial}_0 \circ \widehat{\partial}_0^T \equiv \widehat{L}_0$, $\widehat{L}_1(\mathcal{M}_1) = \widehat{\partial}_1^T \circ \widehat{\partial}_1 \neq \widehat{L}_1$.

Combinatorial Laplacians can be used to compute various useful topological characteristics of a PCC Mieghem (2010); Weber et al. (2017). One example is the well-known Betti numbers, which are obtained simply by Friedman (1998):

$$\widehat{\beta}_p = \text{Dimension}(\text{Nullspace}(\widehat{L}_p)), \tag{24}$$

i.e., the number of zero eigenvalues of $\widehat{L}_p$. The eigenvalues of $\widehat{L}_p$ are non-negative real numbers, which provide the essential information about the topology formed by the p-cells van Mieghem (2011). These have been used in many areas to characterise complex networks Aktas et al. (2019); Agami and Adler (2020); Wanner et al. (2010).

The Betti numbers are $\beta_0(\mathcal{M}_1) = \beta_0(\mathcal{M}_2)$, which gives the number of connected components of 1-cells and 2-cells; $\beta_1(\mathcal{M}_1)$ which gives the number of one-dimensional holes, i.e., closed cycles of 1-cells (2-cells are not included in $\mathcal{M}_1$); and $\beta_2(\mathcal{M}_2)$ which gives the number of two-dimensional holes, i.e. closed cycles of 2-cells (3-cells are not included in $\mathcal{M}_2$).

The alternating sum of Betti numbers gives the well-known Euler characteristic, which separates topologically different structures:

$$\widehat{\chi} = \sum_{p\in\{0,1,2\}} (-1)^p \widehat{\beta}_p = \widehat{\beta}_0 - \widehat{\beta}_1 + \widehat{\beta}_2. \tag{25}$$

The positive value of $\widehat{\chi} > 0$ corresponds to the structures homotopically equivalent to the sphere; the structures with $\widehat{\chi} = 0$ are equivalent to the torus, and in the last case of $\widehat{\chi} < 0$ corresponds to the most complicated case of several connected tori Spanier (1989).

A derivative topological characteristics is the *inverse connectivity* Zhu et al. (2023a):

$$\widehat{I}_c = \ln\left(\frac{\widehat{\beta}_0}{\widehat{\beta}_2}\right), \tag{26}$$

whose negative values indicate high connectivity in the considered structure.

L_p contains the topological characteristics of the whole p-skeleton, whereas in most cases, only the structures of special p-cells with their topologies are of primary interest. The p-cells of a given special type form a substructure which algebraically is described by reduced incidence matrices ∂_p (see Section 2). The null space of the corresponding reduced Laplacians $L_p = \partial_p^T \cdot \partial_p + \partial_{p+1} \cdot \partial_{p+1}^T$ gives the Betti numbers Friedman (1998):

$$\beta_p = \text{Dimension}\left(\text{Nullspace}\left(L_p\right)\right), \tag{27}$$

and the values of the Euler characteristic χ^p and the inverse conductivity I_c^p corresponding to the structures of special cells placed on the corresponding p-skeletons Zhu et al. (2023a).

The PCC Processing Design code Borodin (2025) can be used for computation of all the discussed topological characteristics for arbitrary PCCs and any given vectors of assigned S_p^a and generated S_p^g substructures of special cells. For these purposes, it uses two external C++ libraries – Eigen Benoit and Gael (2022) and Spectra Qiu (2022), allowing fast calculation of spectra of large, sparse matrices.

5. Metrics and Associated Cell Energies

A PCC, being a purely combinatorial construction Kozlov (2008); Berbatov et al. (2022), has no metric without the explicit *embedding* of $\mathcal{T}^p$ into a physical metric space as $\mathcal{R}^p$, which allows the assignment of 1-cell lengths, 2-cell areas and 3-cell volumes. Correspondingly, no physical energies can be associated with the $\mathcal{T}^p$ cells. There is an analogy with graphs ($\mathcal{M}^1$), which are defined as a family of 2-tuples. Drawing a graph requires an embedding space and nodal coordinates or edge lengths concerning the metric of the embedding space; different metrics can produce completely different drawings of the same graph Bollobas (1998). Fig.23 illustrates this on the example of Manchester (UK) Metrolink graph and its embedding on the 2-dimensional geographical map, preserving relative distances between the stations (nodes).

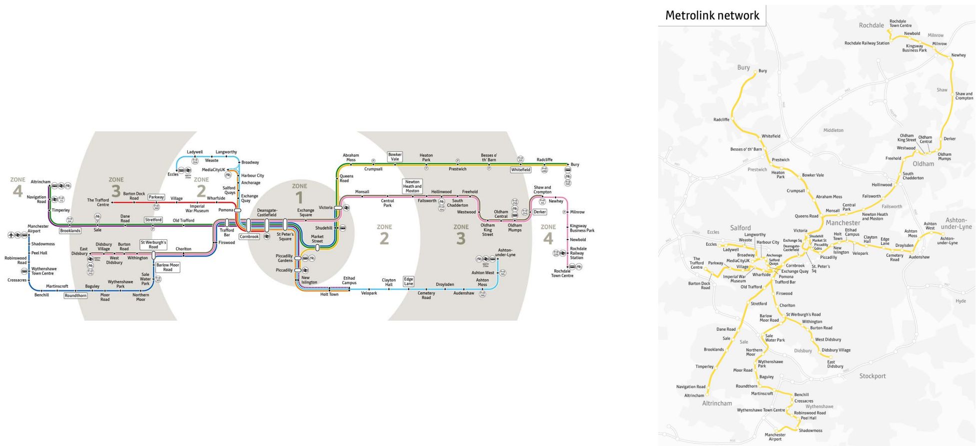


Figure 23: An example of Manchester (UK) Metrolink represented as a graph (left) and its embedding in a 2D geographical map (right).

Although the combinatorial distances (see Zones 1-4 in Fig. 23) can be effectively used even in the graph structure, only real distances allow estimation of the physical characteristics such as travelling time.

5.1. PCC embeddings and introduction of a metric

There is less freedom in the embedding of a $\mathcal{M}^2$ or $\mathcal{M}^3$ in a continuous space compared with a graph, because the number of relations between the cells of different dimensions $p = \{0, 1, 2, 3\}$ is much higher. A default setting for the tessellation in Neper Quey (2021) software is a 1x1x1 cubic volume, which allows a simple scaling of the whole model by assigning $\lambda_1 \neq 1$ as a measure to the linear size $\mathcal{T}^3$ along one of the cube edges. The Neper post-processing options additionally provide an opportunity to output explicitly all lengths, areas and volumes in a tessellation that makes the embedding of a PCC a straightforward procedure once the linear size $\lambda_1(\mathcal{T}^3)$ of the tessellation is done. In particular, in a cube with $\lambda_1 = 10\mu$m, tessellated into $N_3 = 10,000$ 3-cells, the average volume of the polyhedra can be estimated roughly as $\overline{d} \approx \sqrt[3]{\lambda_1^3/N_g} \approx 460$ nm. Similarly, assigning a tessellation's face area $\lambda_2 \neq 1$ or its volume $\lambda_3 \neq 1$ provides a similar 'homogeneous' embedding of a PCC in the corresponding metric space. However, 'heterogeneous' embeddings scaling different 3-cells differently according to some distribution of polyhedra volumes or extending one of the PCC's edges more than others are also possible.

5.1.1. Cell energies

The length scale allows for calculating distances between the discrete fields of any scalar or tensorial physical quantities assigned to the barycentres of p-cells in a PCC. Given the sizes and physical parameters of energy densities $\epsilon_i^p = \{\epsilon_1, \epsilon_2, \ldots, \epsilon_n\}_p$ the corresponding to the p-cell energies can be defined in barycentre of each cells as

$$\mathcal{E}_i^p = const \cdot \epsilon_i^p \Lambda_i, \tag{28}$$

where $\Lambda_i = \Lambda_i(\lambda)$ is the corresponding measure equal to the length for 1-cells, areas for 2-cells, and volumes for 3-cells, and the constant can express additional physical effects.

Similarly, any other physical characteristics which can be expressed as a function of energies or potential functions can be defined at the barycentres of p-cells. What is more important, such physical characteristics can also depend on the combinatorial types and indices of the adjacent and incident cells, as well as mean-field Barroso et al. (2025) and 'global' characteristics of the network of special cells.

6. Crystallography

In physical applications, the existence of various low-dimensional 'nanostructures' is often assumed 'inside' the considered discrete set of PCC cells. Atomistic structures of materials are such an important example for the applications of PCC-based modelling in materials science and materials design Bushuev et al. (2025); Zhu et al. (2023b). In particular, the experimental grain orientations can be extracted directly from the Electron Backscatter Diffraction (EBSD) maps obtained by a scanning electron microscopy (SEM) Patala et al. (2012); Wright et al. (2011) using additional software such as MTEX Hielscher (2022).

The set of crystallographic orientations at pixels obtained by the conventional EBSD techniques is not convenient for statistical studies of GBs, as it contains many artefacts. Even the use of additional software like MTEX Hielscher (2022) cannot remove these artefacts entirely. However, for problems where only the topological information about the grains' neighbourhood is essential, most of the geometrical information provided by the noisy data is excessive. Suitable for such studies is a representation based on the coordinates of grains' barycentres and an assumption about their topology. A realistic topological representation of grain structure can be provided by the well-studied Voronoi tessellation Aurenhammer (1991) of 2-dimensional space corresponding to the experimental EBSD maps of the selected materials. Voronoi tessellations can be calculated by several software tools; the free Neper software Quey (2021) allows one to create a Voronoi tessellation with several additional characteristics such as grain sphericity, and texture analysis, including density pole figures. Two experimental data files are needed for creating the tessellations: (1) coordinates of grains' barycentres; and (2) grain orientations expressed in Euler angles. Based on these files, Neper outputs a ".tess" file containing all the necessary topological information encoded. Fig. 24 shows examples of two Neper-based tessellations for microstructures Cu-0.1%Cr-0.1%Zr and Cu-0.3%Cr-0.5%Zr alloys subjected to four passes of equal-channel angular pressing Bodyakova et al. (2022b,a). The Neper output files are used as input to a purpose-built Python code Polyhedral Cell Complex Analyser (VCA code) Bushuev (2023). This code contains classes for all types of structural elements (grains, GBs and GB triple junctions) of the Voronoi tessellation of a plane and provides full topological information on their connectivity and neighbourhood. VCA code translates the Voronoi tessellation into a PCC giving direct access to each element with associated scalar and/or vector values, such as orientations of grains (2-cells in $\mathcal{M}^2$) and disorientations at GBs (1-cells in $\mathcal{M}^2$).

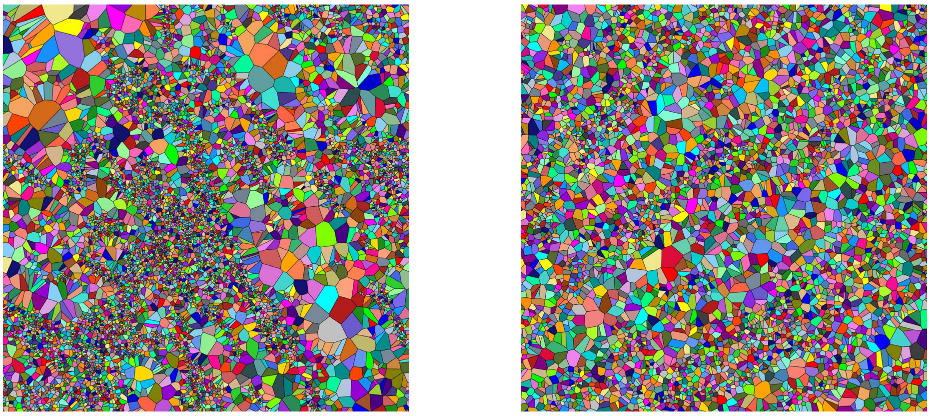


Figure 24: Two examples of 2D Voronoi tessellations provided by NeperQuey (2021) based on the experimental data (barycenter coordinates and Euler angles) for grain structures of Cu-0.1%Cr-0.1%Zr and Cu-0.3%Cr-0.5%Zr alloys subjected to 4 ECAP passes Bushuev et al. (2025).

6.1. Texture representation

Texture refers to the existence of different spatial orientations of the crystalline lattice in different constituents of a polycrystalline assembly. Electron microscopy techniques, such as electron backscatter diffraction (EBSD), are typically used to measure lattice orientations and obtain orientation distribution functions (ODF). EBSD with high spatial resolution can provide an ODF of nanometre-scale microstructures Randle (2009). The ODF typically affects significantly macroscopic material properties, and the prediction of its evolution during materials' processing still remains a challenging task Fundenberger and Beausir (2023).

A typical EBSD measurement provides the orientation of the crystalline lattice at each spatial position with respect to a fixed external coordinate system in terms of Euler angles (ϕ_1, Φ, ϕ_2). These can be converted to unit quaternions, $q = (q_0, q_1, q_2, q_3)$, which are more computationally efficient and avoid the singularity at $\Phi = \pi/2$ Morawiec (2003); Kobayashi and Warren (2005). The vector part (g_1, g_2, g_3) is a rotation axis – a vector with respect to the external coordinate system. The scalar part is $g_0 = cos(\theta/2)$, where θ is the angle of rotation, by which the local lattice triad needs to be rotated around the axis to coincide with the external coordinate system.

Since the components of a unit quaternion are related by:

$$g_0^2 + g_1^2 + g_2^2 + g_3^2 = 1, \tag{29}$$

the orientations of all crystallites in a material can be represented as points in a quaternion space with coordinates $\{Q_1, Q_2, Q_3\}$ given by:

$$Q_1 = g_1^2/(1-g_0^2),\ Q_2 = g_2^2/(1-g_0^2),\ Q_3 = g_3^2/(1-g_0^2), \tag{30}$$

with $Q_1 + Q_2 + Q_3 = 1$. Such representation allows for visualising changes of grain orientations as displacements of points within the quaternion triangular space along the mesh lines (see Fig. 25).

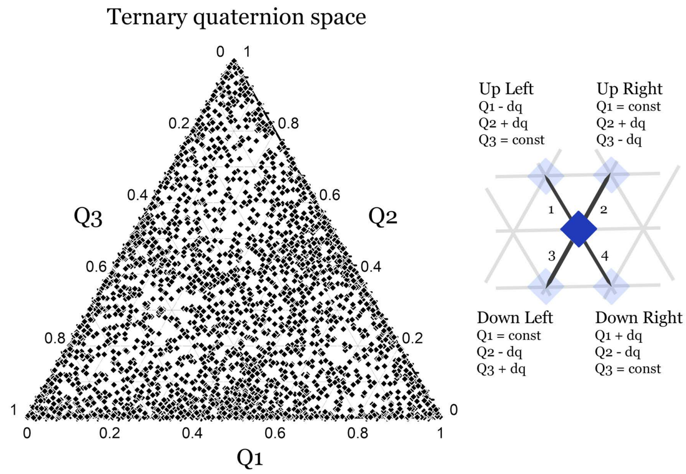


Figure 25: Ternary space of unit quaternions (left), where each point corresponds to grain orientation; and possible rotations (dq shifts) of a quaternion axis as translations to neighbouring mesh nodes (right).

6.2. Grain boundary disorientations and long-range disorientation distributions

Generally, the ODF can be incorporated into a polycrystalline assembly by assigning crystallographic orientations to all grains (volumes of the corresponding polyhedral assembly). The misorientation between two adjacent grains or between two sub-grains of a single grain is the rotation required to transform a tensorial quantity, such as stress or strain, from one set of crystal axes to the other Zhu et al. (1999); Engler and Randle (2009). The misorientation that transforms quantities from grain A to grain B can be calculated by $\Delta g_{AB} = g_B \circ g_A^{-1}$, where g_A^{-1} is the inverse quaternion. This is illustrated in Fig. 26. Generally, the rotation angle of the quaternion Δg_{AB} is in the interval $[0, \pi]$. However, the presence of crystal symmetry reduces the possible rotation angles. For example, Face Centred Cubic (FCC) crystals have 24 symmetrically related orientations, which are physically indistinguishable, but mathematically distinct. The crystal symmetries are represented by

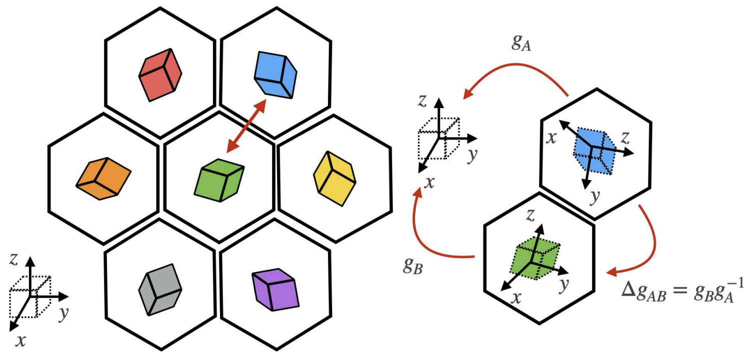


Figure 26: Orientations of crystallographic axes in different structural elements with respect to an external reference frame; Misorientation describing transformation between crystal reference frames across a boundary element Zhu et al. (2023a).

crystal symmetry operators S, so that all possible misorientations are given by the quaternions Mackenzie (1958):

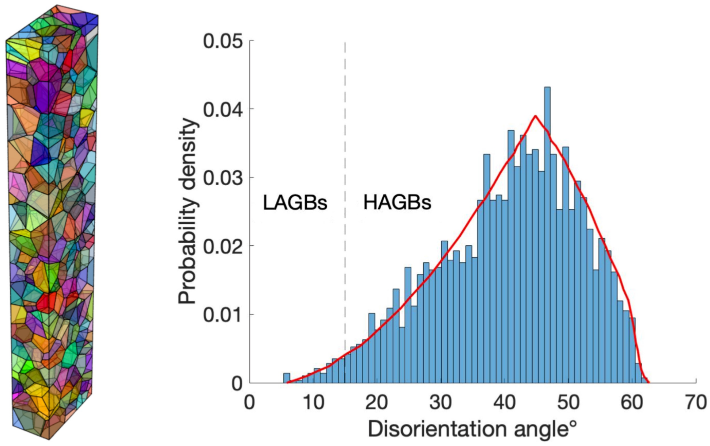
$$\Delta g_{AB} = S_B \circ g_B \circ (S_A \circ g_A)^{-1}. \tag{31}$$

$\Delta g = \Delta g_{AB}$ is a rotation matrix and can also be represented compactly with a unit quaternion $\{q_0, q_1, q_2, q_3\}$ with the smallest rotation angle satisfying the condition that the rotation axis falls within the standard stereographic triangle associated with the relevant point group, and $q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1$ Patala et al. (2012), where $q_0 = cos(2\theta)$ with θ being the angle of rotation, and $\{q_1, q_2, q_3\}$ is the axis of rotation. The quaternion components are related to the components r_{ij} of Δg by:

$$q_0 = \frac{\sqrt{1 + r_{11} + r_{22} + r_{33}}}{2}, \quad q_1 = \frac{r_{32} - r_{23}}{4q_0}, \quad q_2 = \frac{r_{13} - r_{31}}{4q_0}, \quad q_3 = \frac{r_{21} - r_{12}}{4q_0}. \tag{32}$$

Disorientation quaternions and their angles of rotation can be calculated for each grain boundary (edge in $\mathcal{M}^2$) as well as for each pair of grains in the PCC. We will use the term disorientation as a substitute for the angle of rotation henceforth.

A map of orientations to all grains/sub-grains provides complete system information, from which the disorientation distribution (DDF) at grain boundaries can be calculated by Eq. (31). An example of disorientation angle distribution is shown in Fig. 27. The calculated disorientation quaternions between adjacent grains can be assigned to their common boundary in the polycrystalline material (common face in the corresponding polyhedral assembly). However, the information provided by disorientation quaternions



(a)

(b)

Figure 27: (a) Polycrystalline assembly of 500 grains with randomly assigned lattice orientations; (b) Calculated disorientation angle distribution (bar chart) and distribution for random texture from Mackenzie (1958) (curve) Zhu et al. (2023a).

is too complex for the purposes of topological analysis. It is practical to consider a simplified model with a discrete set of grain boundary types. In such a way, the special 2-cell types in the PCC can also be *induced* by the crystallographic (including experimental EBSD maps) data.

The research area of grain boundary engineering commonly classify the boundaries into discrete sets on the basis of the coincident site lattice model Johnson et al. (2018), which is one effective classification of GB networks, however a binary classification Frary and Schuh (2005); Borodin and Jivkov (2019); Zhu et al. (2021) separating grain boundaries into low-angle grain boundaries (LAGBs) (with disorientation angles less than 15 degrees), and high-angle grain boundaries (HAGBs) (with larger disorientation angles) can be equally introduced Borodin and Jivkov (2019); Zhu et al. (2021); Bushuev et al. (2025).

6.2.1. Random Mackenzie distribution of disorientations

The disorientation angle distribution function (DDF) for random grain orientations is a benchmark in the studies of correlations between any particular set of grain orientations. It was derived in the classical work Mackenzie (1958) by Mackenzie. An important characteristic of this distribution is the low fraction of LADs or LAGBs ($\theta \leq 15$ degrees) equal to 0.02277. This analytical distribution assumes an infinite number of grains. A random (homogeneous) distribution of grain orientations can be generated by Neper Quey (2021). In this case, the number of grains is finite, and the resulting disorientation distribution fluctuates around the theoretical Mackenzie distribution with a magnitude strongly dependent on the number of elements in the PCC. As it is well-known from statistics, this magnitude scales with $1/\sqrt{N}$, where N is the number of disorientations.

In addition to the commonly used DDF, let us consider the kth disorientation distribution function (k-DDF) calculated from the disorientations between the grains and all their k-neighbours (see Fig. 10). Together with the k-DDF histograms, Hellinger distances can also be calculated Kitsos and Toulias (2017) for different numbers of disorientation angle bins – from 2 to 32 (see Fig. 22). In the case of two bins, the first bin contains disorientations with angles $\theta < 15$ (LADs) and the second bin contains disorientations with angles $\theta > 15$ (HADs).

Further Reading

More up-to-date references can be found in the Publications list of the MTERiA project webpage, including:

1. Oleg Bushuev, Elijah Borodin, Anna Bodyakova, Siying Zhu, Andrey P. Jivkov (2025) Disorientation-based classification of mesostructures in severely deformed copper alloys. Acta Materialia, 286, 120714. doi: 10.1016/j.actamat.2025.120714

2. Elijah Borodin and Afonso D. M. Barroso, Andrey P. Jivkov (2024) Kinetic view on dynamic plasticity of crystalline solids. arXiv. doi: 10.48550/arXiv.2412.14105

3. Elijah Borodin, Oleg Bushuev, Vladimir Bratov, Andrey P. Jivkov (2024) Discrete model for discontinuous dynamic recrystallisation applied to grain structure evolution inside adiabatic shear bands. Journal of Materials Research and Technology, 30, 2125-2139. doi: 10.1016/j.jmrt.2024.03.206

4. E.N. Borodin, A.G. Sheinerman, O.Yu. Bushuev, M.Yu. Gutkin, A.P. Jivkov (2024) Defect-induced fracture topologies in Al2O3 ceramic-graphene nanocomposites. Materials & Design, 239, 112783. doi: 10.1016/j.matdes.2024.112783

5. Siying Zhu, Elijah Borodin, Andrey P. Jivkov (2024) Discrete modelling of continuous dynamic recrystallisation by modified Metropolis algorithm. Computational Materials Science, 234, 112804. doi: 10.1016/j.commatsci.2024.112804

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7. Siying Zhu, Elijah Borodin, Andrey P. Jivkov (2023) Topological characteristics of grain boundary networks during severe plastic deformations of copper alloys. Acta Materialia, 259, 119290. doi: 10.1016/j.actamat.2023.119290

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doi: 10.1016/j.ijsolstr.2021.111345

10. E.N. Borodin, A.P. Jivkov, A.G. Sheinerman, M.Yu. Gutkin (2021) Optimisation of rGO-enriched nanoceramics by combinatorial analysis. Materials & Design, 212, 110191.
doi: 10.1016/j.matdes.2021.110191

11. S. Zhu, E.N. Borodin, A.P. Jivkov (2021) Triple junctions network as the key structure for characterisation of SPD processed copper alloys. Materials & Design, 198(24), 109352.
doi: 10.1016/j.matdes.2020.109352

12. D. Seruga, O. Kosmas, A.P. Jivkov (2020) Geometric modelling of elastic and elastic-plastic solids by separation of deformation energy and Prandtl operators. International Journal of Solids and Structures, 198, 136–148.
doi: 10.1016/j.ijsolstr.2020.04.019

13. E. N. Borodin, A. P. Jivkov (2019) Evolution of triple junctions' network during severe plastic deformation of copper alloys – a discrete stochastic modelling. Philosophical Magazine, 100(4), 467-485.
doi: 10.1080/14786435.2019.1695071

14. E.N. Borodin, A. Morozova, V. Bratov, A. Belyakov, A.P. Jivkov (2019) Experimental and numerical analyses of microstructure evolution of Cu-Cr-Zr alloys during severe plastic deformation. Materials Characterization, 156, 109849.
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